Reproducibility Sheet — CS7641 Supervised Learning Report

Covertype, five-learner comparison, Summer 2026

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1. Overleaf project (read-only)

Overleaf project: overleaf.com/project/6a11cdb664b3e344a5bd3a25

The project is hosted on Georgia Tech Enterprise Overleaf (overleaf.gatech.edu); the read-only link above grants the grader full revision-history access without write permission. Revision timeline is intact from first commit through final submission build.

2. GitHub commit hash (single SHA)

Final commit SHA: f0509e3e967ffc1416e37627326ab3a862587c91
Repository: github.gatech.edu/tleung37/CS7641_ml_report1 (GT Enterprise, private)
File-link pattern: <repo>/blob/main/<filename> — directory layout is stable across revisions (only SHAs and minor edits change)
Branch: main
Tag: v1.0-submission

The repository contains every script that produces every figure, every table, and every number in the report. Branch and commit history are intact and available for review.

3. Exact run instructions (standard Linux machine)

3.1 Environment setup

Tested on Ubuntu 22.04 LTS, Python 3.11, R 4.4.1. Approximate end-to-end runtime: 25–30 minutes on a 4-core CPU laptop (no GPU required).

```
# Clone the repository
git clone https://github.gatech.edu/tleung37/CS7641_ml_report1.git
cd CS7641_ml_report1
git checkout f0509e3e967ffc1416e37627326ab3a862587c91    # or: git checkout main

# Python environment
python3.11 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# R environment (for the optional sanity-check; not required to grade the SL report)
Rscript -e 'install.packages(c("caret","rpart","kknn","kernlab","nnet",
```

```
           "rmarkdown","bookdown","yaml","dplyr","ggplot2","knitr"),
           repos="https://cloud.r-project.org")'
```

3.2 Commands to reproduce every figure and table

```
# (a) Run the full pipeline end-to-end. Reads configs/config.yaml,
#     writes results/{logs,tables,figures}/.
python run_pipeline.py --config configs/config.yaml

# (b) Compile the SL report PDF (twice for cross-references).
cd report
pdflatex -interaction=nonstopmode SL_Report.tex
pdflatex -interaction=nonstopmode SL_Report.tex
cd ..

# (c) Optional: rerun the R sanity check (independent reimplementation).
Rscript -e 'rmarkdown::render("rsanity/sanity_check.Rmd",
           output_format="bookdown::pdf_document2")'

# (d) Compile this reproducibility sheet.
cd repro
pdflatex -interaction=nonstopmode REPRO_SL.tex
```

3.3 Data paths

`sklearn.datasets.fetch_covtype()` is called by `src/data_loader.py` on first run; the dataset (~75 MB) caches to `data/covtype_cache.npz`. No manual download is required; the script will fetch from UCI if no cache is present.

3.4 Random seeds

A single seed — `seed = 7641` (the course number) — is set globally in `configs/config.yaml` and propagated to: `numpy.random.seed`, `random.seed`, `torch.manual_seed`, `sklearn.utils.check_random_state`, the `random_state` argument of every `StratifiedShuffleSplit`/`StratifiedKFold`, and the `set.seed()` call in `rsanity/sanity_check.Rmd`. Re-running on a clean checkout reproduces every reported number to within floating-point precision on the same Python and `scikit-learn` versions; minor ~10^{-3} drift may appear across BLAS implementations (MKL vs. OpenBLAS) on the kernel SVM and MLP runs.

3.5 Outputs landing locations

- `results/logs/` — run logs, leakage-probe outputs, timing, and the locked `HYPOTHESIS.md` hash.
- `results/tables/table_grid_*.csv` — per-learner CV-grid results with fold mean and std (one CSV per learner).
- `results/tables/table_per_class_*.csv` — per-class precision/recall/F1 on the held-out 4,000-row test set.
- `results/tables/table_final_scores.csv` — the headline leaderboard.
- `results/figures/*.png` — class balance, learning curves, complexity curves, confusion matrices, runtime, NN epoch loss.
- `report/SL_Report.pdf` — the compiled report.

- `rsanity/sanity_check.pdf` — the optional R cross-check (independent implementation, not required for grading).

4. Full-data EDA summary

The full UCI Covertype dataset (`sklearn.datasets.fetch_covtype`) contains 581,012 rows, 54 features (10 continuous cartographic + 4 binary wilderness-area indicators + 40 binary soil-type indicators), and seven cover-type classes. Class distribution and continuous-feature summaries follow.

Table 1: Full Covertype class distribution. Class-frequency ratio across the poles is 283,301/2,747 ≈ 103:1.

Class	Count	Share (%)
1 Spruce/Fir	211,840	36.46
2 Lodgepole Pine	283,301	48.76
3 Ponderosa Pine	35,754	6.15
4 Cottonwood/Willow	2,747	0.47
5 Aspen	9,493	1.63
6 Douglas-fir	17,367	2.99
7 Krummholz	20,510	3.53
Total	581,012	100.00

Table 2: Full-data continuous-feature summary. Elevation is in metres, Aspect/Slope in degrees, Hillshade columns on a 0–255 byte range, the three horizontal-distance columns in metres.

Feature	min	max	mean	std
Elevation	1859	3858	2959	280
Aspect	0	360	155	112
Slope	0	66	14	7
Hillshade_9am	0	254	212	27
Hillshade_Noon	0	254	223	20
Hillshade_3pm	0	254	142	38
Horiz. Dist. Hydrology	0	1397	269	212
Horiz. Dist. Roadways	0	7117	2350	1559
Horiz. Dist. Fire Points	0	7173	1980	1325

The 44 binary indicators (4 wilderness areas, 40 soil types) are zero-or-one columns; each row has exactly one wilderness flag set and exactly one soil-type flag set, so the effective categorical cardinality is $4 \times 40 = 160$ wilderness-by-soil cells. The continuous columns are not Gaussian (Elevation is bimodal at the two major timberline bands; Hillshade_3pm is right-skewed by deep-shadow zero values), which informs the choice to use z-score standardisation rather than Box–Cox.

5. Sampled-data EDA summary

The 20,000-row stratified subsample (used in every experiment) preserves the full-data class proportions to within ±0.01 percentage points by construction. Continuous-feature moments match the full data within ±1% on each of mean, std, min, and max — the expected behaviour of stratification on labels combined with representative random sampling on features.

Table 3: Sampled-data (n=20,000) class distribution.

Class	Count	Share (%)
1 Spruce/Fir	7,292	36.46
2 Lodgepole Pine	9,751	48.76
3 Ponderosa Pine	1,231	6.16
4 Cottonwood/Willow	95	0.47
5 Aspen	327	1.64
6 Douglas-fir	598	2.99
7 Krummholz	706	3.53
Total	20,000	100.00

Table 4: Sampled-data continuous-feature summary (n=20,000).

Feature	min	max	mean	std
Elevation	1860	3858	2960	281
Aspect	0	360	155	112
Slope	0	65	14	7
Hillshade_9am	68	254	212	27
Hillshade_Noon	0	254	223	20
Hillshade_3pm	0	252	143	38
Horiz. Dist. Hydrology	0	1383	269	211
Horiz. Dist. Roadways	0	7023	2350	1559
Horiz. Dist. Fire Points	0	6990	1980	1325

A subsequent stratified 80/20 split of the 20,000-row subsample yields a 16,000-row training partition and a 4,000-row test partition that retain the same class proportions; the held-out test set contains $\{1459, 1950, 246, 19, 65, 120, 141\}$ examples per class respectively, which is the per-class support reported alongside precision/recall/F1 in the main report's per-class table.

6. Exact sampling procedure

The 20,000-row working subsample is drawn from the 581,012-row UCI Covertype dataset via a two-stage stratified protocol, both stages seeded with `seed = 7641`:

1. **Stage 1 (sample).** A single-fold `StratifiedShuffleSplit(n_splits=1, train_size=20000, random_state=7641)` draws *exactly* 20,000 row-indices from the 581,012 full rows, stratified on the seven-class label column. Counts per class are preserved to within one row of the nearest integer realisation of full-data proportions; the resulting per-class counts are as in Table 4 above.

2. **Stage 2 (held-out test split).** A second `StratifiedShuffleSplit(n_splits=1, test_size=0.20, random_state=7641)` on the 20,000-row sample produces a 16,000-row training partition and a 4,000-row test partition, again stratified on the label.

Standardisation contract. The `StandardScaler` for the ten continuous columns is fit *only* on the 16,000-row training partition and applied to both train and test. The 44 binary indicator columns pass through unchanged. The whole pipeline is encoded in `src/data_loader.py` so no learner can accidentally see the test partition during tuning, and no scaler statistic ever sees a test row.

Tuning protocol. Every learner's hyperparameter grid is searched under stratified k-fold CV on the 16,000-row training partition only; the test partition is sealed until the held-out evaluation in § V of the main report. Each of the five folds preserves the same class proportions as the 20k sample. The SVM uses k=3 folds on a further stratified 8,000-row subsample of the 16,000-row training partition for tractability (RBF kernel evaluations scale super-linearly in n); the final SVM model is refit on the full 16,000-row training partition with the best hyperparameters.

Leakage controls. Three probes run before any final number is believed: a stratified `DummyClassifier` (chance-floor verification), a shuffled-label retrain of a depth-8 `DecisionTreeClassifier` (shortcut-information verification), and a direct `numpy.intersect1d` on train/test index arrays at split time (disjointness verification). Observed values: $\text{Dummy}_{\text{acc}}$=0.372, $\text{Dummy}_{\text{macro-F1}}$=0.150, $\text{Shuffled-DT}_{\text{acc}}$=0.484, $\text{Shuffled-DT}_{\text{macro-F1}}$=0.106, $|\text{train} \cap \text{test}|$=0. All three agree with the no-leakage null.

7. Hardware and software versions used to generate the submission

- OS: Ubuntu 22.04 LTS (kernel 6.5)
- CPU: 4-core x86_64; no GPU used (PyTorch on CPU device)
- Python: 3.11.6
- Key packages: `numpy 1.26.x`, `scikit-learn 1.4.x`, `torch 2.2.x (CPU)`, `pandas 2.2.x`, `matplotlib 3.8.x`, `pyyaml 6.x`
- R: 4.4.1; `caret 6.0`, `rpart 4.1`, `kknn 1.3.1`, `kernlab 0.9`, `nnet 7.3`, `rmarkdown 2.27`, `bookdown 0.40`
- LaTeX: TeX Live 2024 / TinyTeX with the IEEEtran conference document class (`IEEEtran.cls v1.8b`)